

DriVR: Extending Driver Training for Persons with Disabilities

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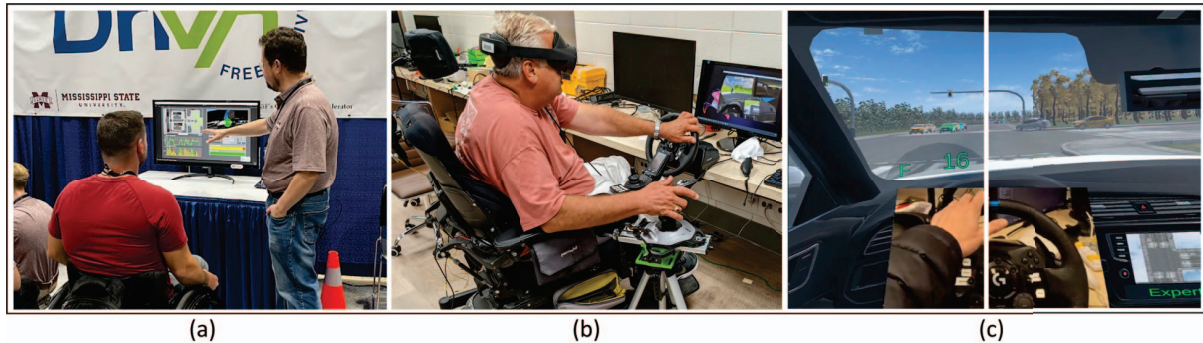


Figure 1: (a) CDRS interface for assessing user performance with data provided by the VR training system, (b) a wheelchair user interacting with a simulated assistive driving technology, (c) a screen capture of the user's view of the mixed reality steering wheel inside of a virtual car width side-by-side view of motion sickness mitigation.

ABSTRACT

DriVR is a prototype driver training tool for at-home use by persons with disabilities. Comprised of off-the-shelf components, DriVR strives to minimize barriers to entry for people in need of training with assistive technology for driving. In this research demonstration, we introduce our prototype and outline some of its key technical features.

Index Terms: [Social and Professional Topics]: Professional Topics—Computing Profession - Assistive Technologies; [Human-Centered Computing]: Human-Computer Interaction (HCI)—Interaction Paradigms - Virtual Reality

1 INTRODUCTION

Driving is an ability that is often taken for granted. However, for many people with disabilities, their freedom and independence can be constrained by an inability to drive. This particularly holds in rural areas of the United States, where there is little to no access to public transportation. Oftentimes, these individuals must rely on the availability of their families and friends to transport them for employment, religious services, medical appointments, and various other everyday activities. Some of these individuals may possess the

capability to drive vehicles with the aid of assistive technologies that support their specific abilities. Before their vehicles can be equipped with these technologies, they need certification to drive a modified vehicle. Training centers capable of performing this certification are not always easily accessible to populations who do not already have some way to get to them. This issue is further exacerbated by the scarcity of public transportation and sparseness of centers in rural areas.

In this research demonstration we show DriVR, our prototype virtual reality (VR) tool for addressing these problems. DriVR was developed under the National Science Foundation's Convergence Accelerator program. This program aims to support highly interdisciplinary teams in the pursuit of solutions to national-scale societal problems. The program track supporting this work focuses on enhancing opportunities for persons with disabilities. DriVR aims to leverage the widespread availability of virtual reality systems and gaming peripherals to provide at-home training opportunities for persons with disabilities. The reliance on primarily off-the-shelf components is crucial, especially considering individuals with disabilities who may lack the means to drive and are often on a fixed income.

2 SYSTEM OVERVIEW

DriVR serves two vital purposes: 1) providing persons with disabilities the opportunity to practice driving with the aid of assistive technology without the need to travel to training centers and 2) providing data on their progress and training needs to a Certified Driver

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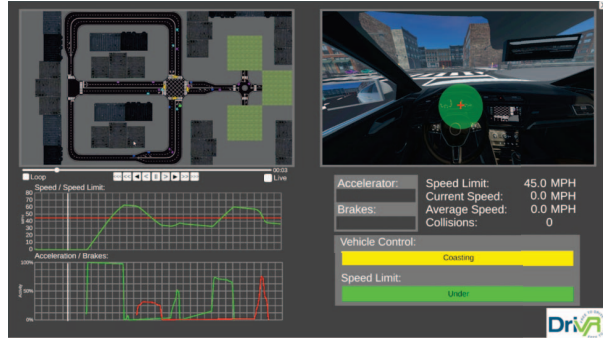


Figure 2: A screen capture of the analytics interface for use by a CDRS in evaluating user performance.

Rehabilitation Specialist (CDRS). While DriVR shares similarities with a video game, it functions as a clinical tool for training and evaluation purposes and has an accompanying data analytics tool to enable a CDRS to monitor progress, see Figure 1a. As part of the development process, we solicited feedback at all stages from potential stakeholders including persons with disabilities (with and without prior driving experience), driving rehabilitation professionals, educational psychologists, and industrial leaders in assistive technologies.

Our prototype utilizes the Meta Quest family of headsets, primarily the Quest Pro and 3, but it could be easily extended to accommodate most off-the-shelf headsets. We also use a Logitech G29 Racing Wheel and Pedals to represent a standard vehicle interface and a Logitech G Extreme 3D Pro joystick to simulate hand-based assistive driving interfaces, see Figure 1b. A crucial aspect of training dexterous tasks involves seeing your hands while performing activities. Despite the increasing sophistication of video-based hand tracking in recent years, accurately and reliably aligning virtual hands with the real-world remains challenging. To remove the need to track hands, we utilized the Quest's mixed reality features, providing a video pass-through view of the user's actual hands and the steering wheel, see Figure 1c.

3 SIMULATOR SICKNESS MITIGATION

Simulator sickness is a common problem seen by VR users and is arguably one of the biggest challenges to its wider adoption. To mitigate this, we have integrated dynamic field of view restriction, see Figure 1c, designed to reduce conflicting motion cues as the user drives the vehicle through the virtual world [1]. Our system offers three field of view modes: Full Field of View (Full), Reduced Field of View (Reduced), and Dynamic Field of View (Dynamic). In Full mode no field of view restriction is applied to the user's view of the world. This mode is primarily designed for experienced VR users who may not be as susceptible to simulator sickness. In Reduced mode the field of view is restricted to approximately 50 degrees in order to minimize motion cues in the user's periphery. This mode is intended for people who are most prone to simulator sickness. In Dynamic mode the field of view restrictions automatically adjust based on the user's driving behavior and location within the environment. This mode automatically balances field of view size with the risk of simulator sickness. In particular, our user feedback indicated that motion sickness typically occurred when turning the vehicle. Following this observation, the field of view in Dynamic mode adjusts based on the turning of the vehicle and its speed.

4 CURATED UNREALITY

Since this is a training and evaluation system specifically for persons with disabilities, there are certain considerations that we needed

to take into consideration. Firstly, during training for any activity, mistakes are inevitable. When driving a real vehicle there could be major consequences to mistakes, such as having a collision. Fortunately, in VR, the physical consequences of a collision are typically minimal. However, VR can evoke emotional responses similar to those in the real world. To avoid increasing user's anxiety towards driving and potentially discouraging them from further training, we introduced some elements to the environment that are intentionally less realistic. There were specifically two elements that we wanted to be unrealistic. The first element involved pedestrians. Many driving situations involve navigating streets or parking lots where pedestrians may also be present. We made all pedestrians resemble black and white illustrations drawn on a cardboard cut mounted to a rolling platform. If a collision occurred between the vehicle and a pedestrian, the cut-out would simply fall over and then spring up again after a brief pause. The second unrealistic component of DriVR involves the physics of vehicular collisions. We adjusted the physics engine such that high speed collisions would not result in catastrophic outcomes. This is especially important as some users may have been previously involved in significant vehicular accidents. Both these modifications aim to provide the user with a sense of safety and comfort during driver training.

5 CDRS ANALYTICS

An essential component of DriVR is its analytics interface, allowing a CDRS to monitor and evaluate the user's performance over time. The analytics interface, shown in Figure 2, allows the CDRS to play back a user's drive through a given environment. In this prototype, we are monitoring collisions, speed, breaking, adherence to posted speed limits, and the location of eye gaze location throughout the drive. The eye tracking data is particularly valuable as it enables the CDRS to determine when and where users are looking during training. This is beneficial for assessing a user's skill in monitoring their instrumentation, blind spots, and mirrors.

6 CONCLUSION

In conclusion, we introduce our prototype driver training system. This prototype was built primarily as a tool to test features and assess effectiveness through continuous stakeholder and user feedback. We envision tools, such as DriVR, becoming an integral part of at-home training and rehabilitation, not only for driving but for numerous other domains as well.

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